STUDENT NAME: EYA GEORGE IBEZIMAKO

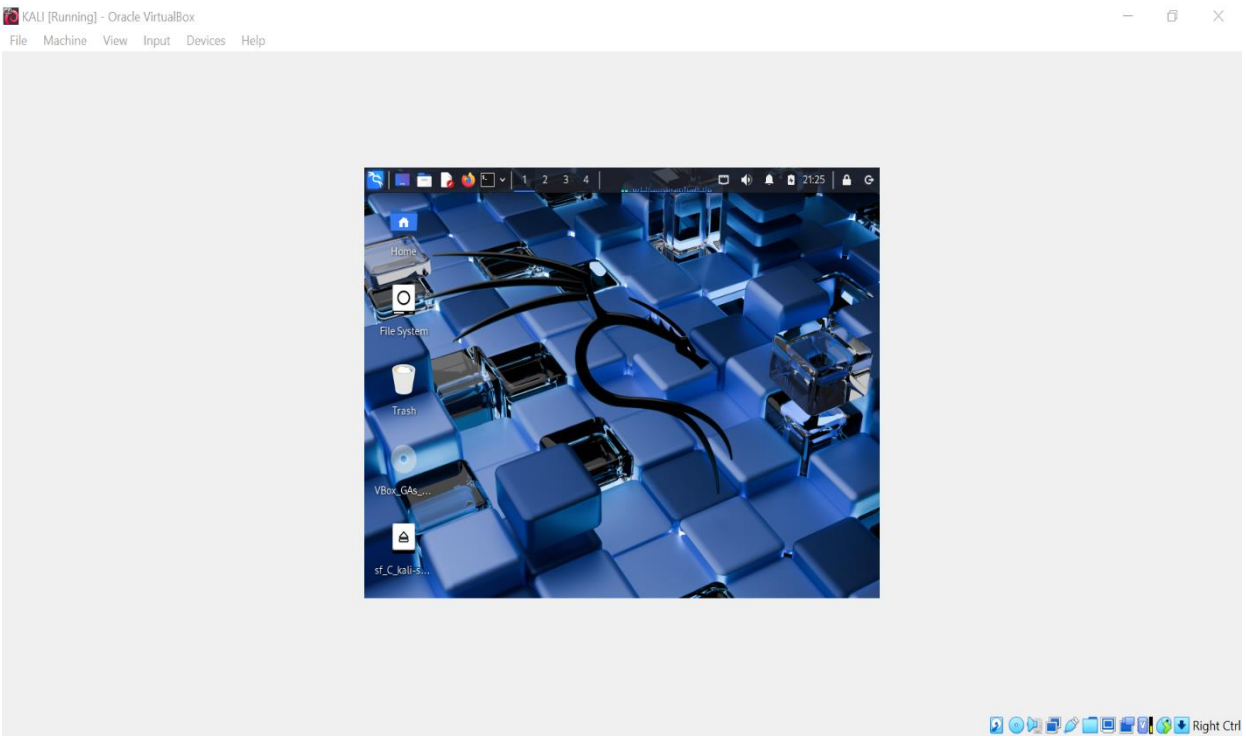
CLASS / COHORT: CYBERSECURITY COHORT 1

DATE SUBMITTED: 10/17/2026

DEADLINE: 10/19/2026

TASK 1: VIRTUAL MACHINE AND OPERATING SYSTEM BASICS

Kali Linux running inside Oracle VirtualBox



Host OS

The Host OS is just the landlord OS installed on the physical computer. In my case, it is Microsoft Windows, which runs on my Dell laptop and provides the environment for running other applications, including VirtualBox.

Guest OS

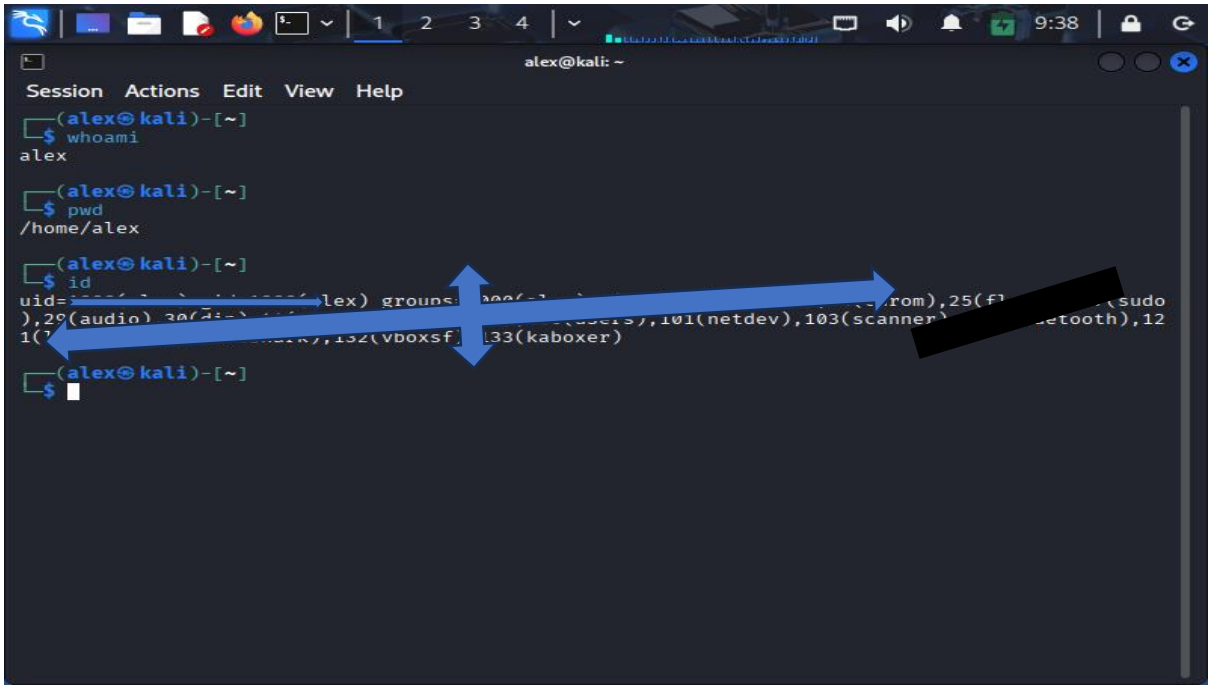
The Guest OS is the OS running inside the virtual machine. In my setup, Kali Linux is the Guest OS. It operates within VirtualBox on my Windows computer and provides a Linux environment for practicing cybersecurity tasks.

VM

A Virtual Machine is a software created computer that runs inside a physical computer. Oracle VirtualBox allows me to run Kali Linux as though it were a separate computer while still using my Windows laptop. It provides a controlled environment for learning Linux commands and practicing cybersecurity skills.

TASK 2: IDENTIFY YOURSELF AND YOUR CURRENT LOCATION

Linux user identity and current filesystem location



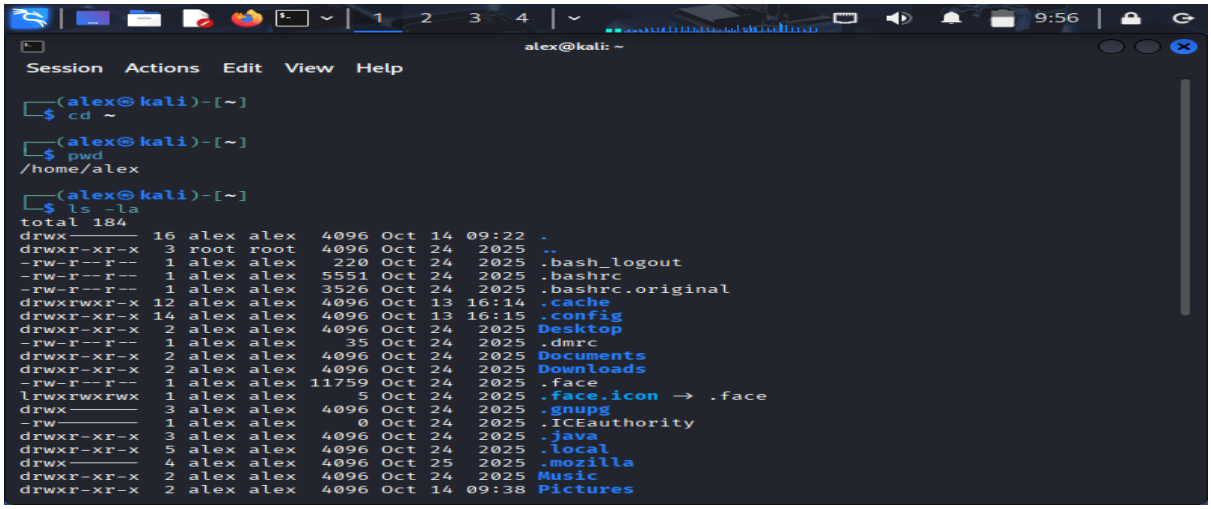
Explanation of Commands and Results

The **whoami** command identifies the username of the account currently logged in to the Linux terminal. In my Kali Linux environment, it displays the username associated with my current session.

The **pwd** command means **"print working directory."** It displays the full path of the directory in which I am currently working in the Linux filesystem.

The id command provides additional information about my user account, including my user **ID (UID),** primary group **ID (GID),** and group memberships. This information helps me understand my account's identity and group associations.

Exploring the Linux Filesystem



Commands used

cd ~

pwd

ls -la

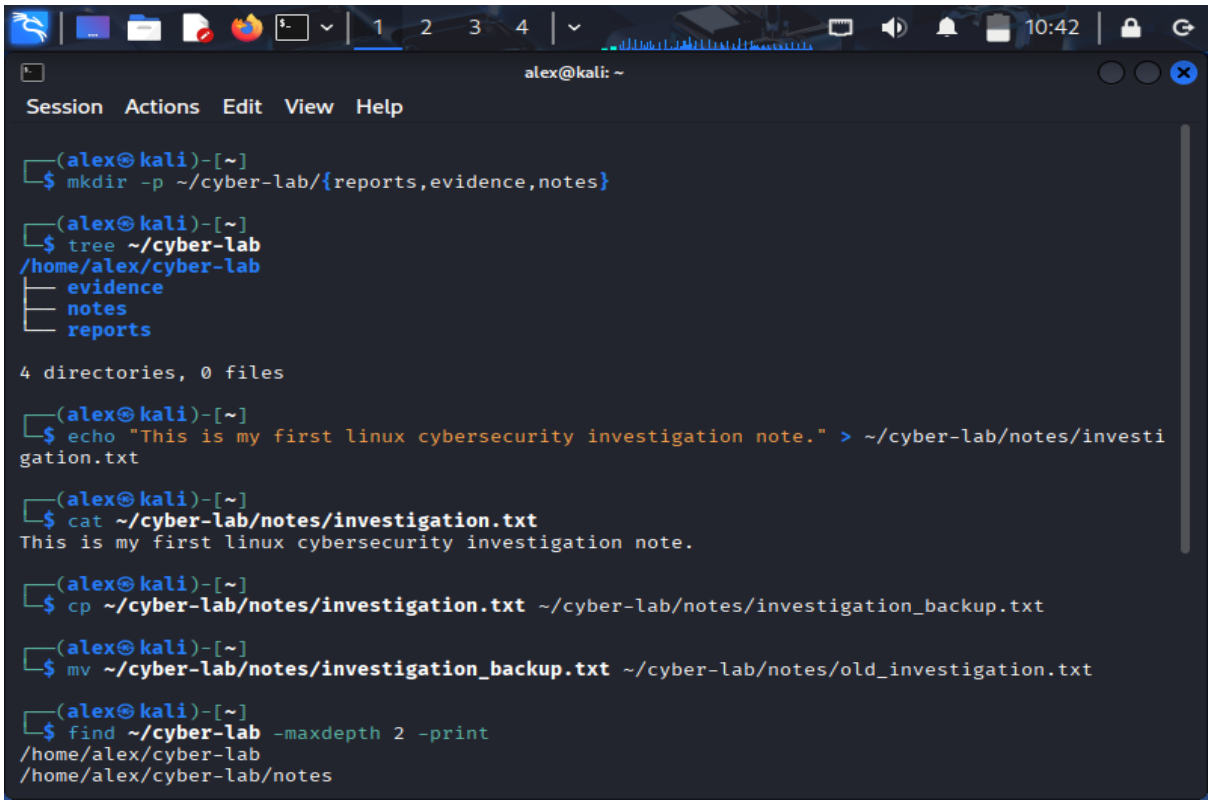
The **pwd** command displayed the path of my home directory. The **ls -la** command listed both ordinary and hidden files and directories in that location.

Identified hidden item: .bashrc

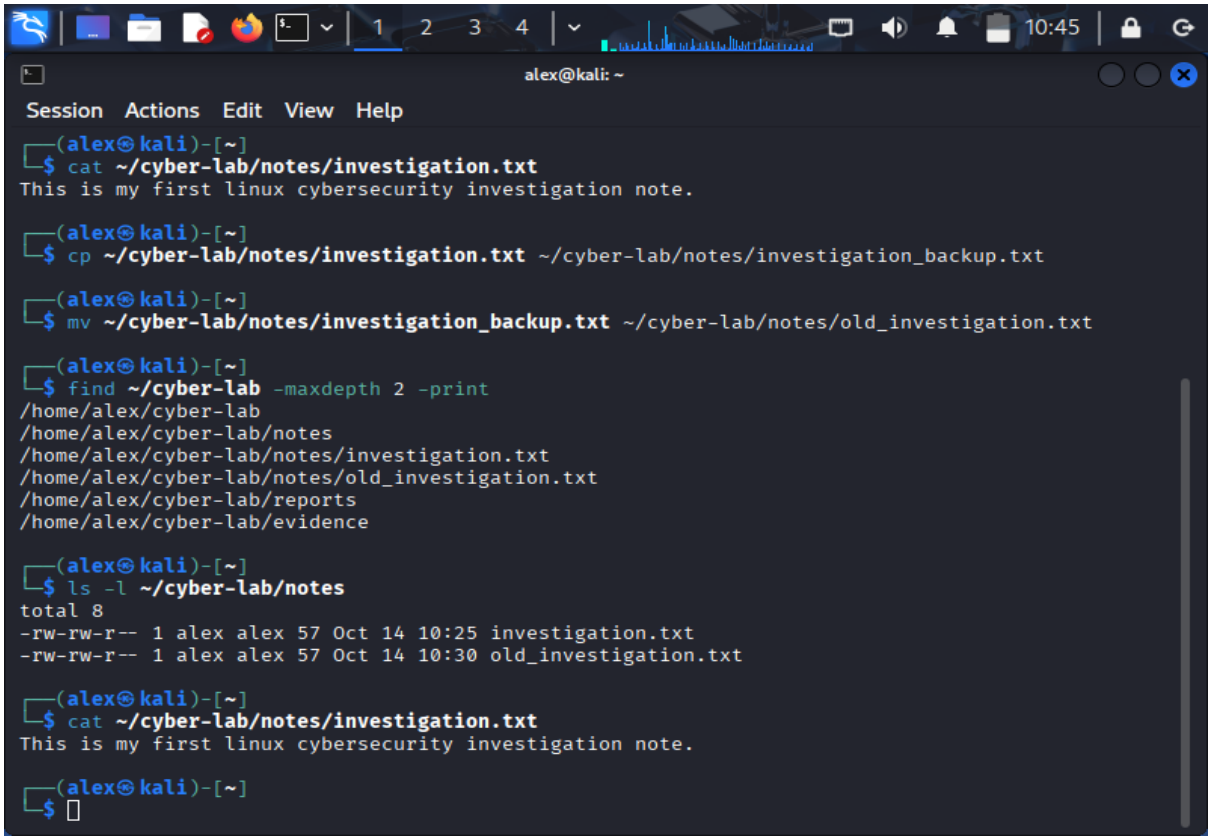
Explanation: I recognized **.bashrc** as a hidden file because its name begins with a dot **(.)**. Linux normally hides filenames beginning with a dot from the standard ls listing. The **-a** option in **ls -la** displays these hidden entries along with ordinary contents.

TASK 4: BUILD A CYBERSECURITY WORKSPACE

Cybersecurity workspace folder structure



Contents of the original investigation note



Commands Used and Explanation

The **mkdir** command was used to create the main cyber-lab directory and its three subdirectories: reports, evidence, and notes.

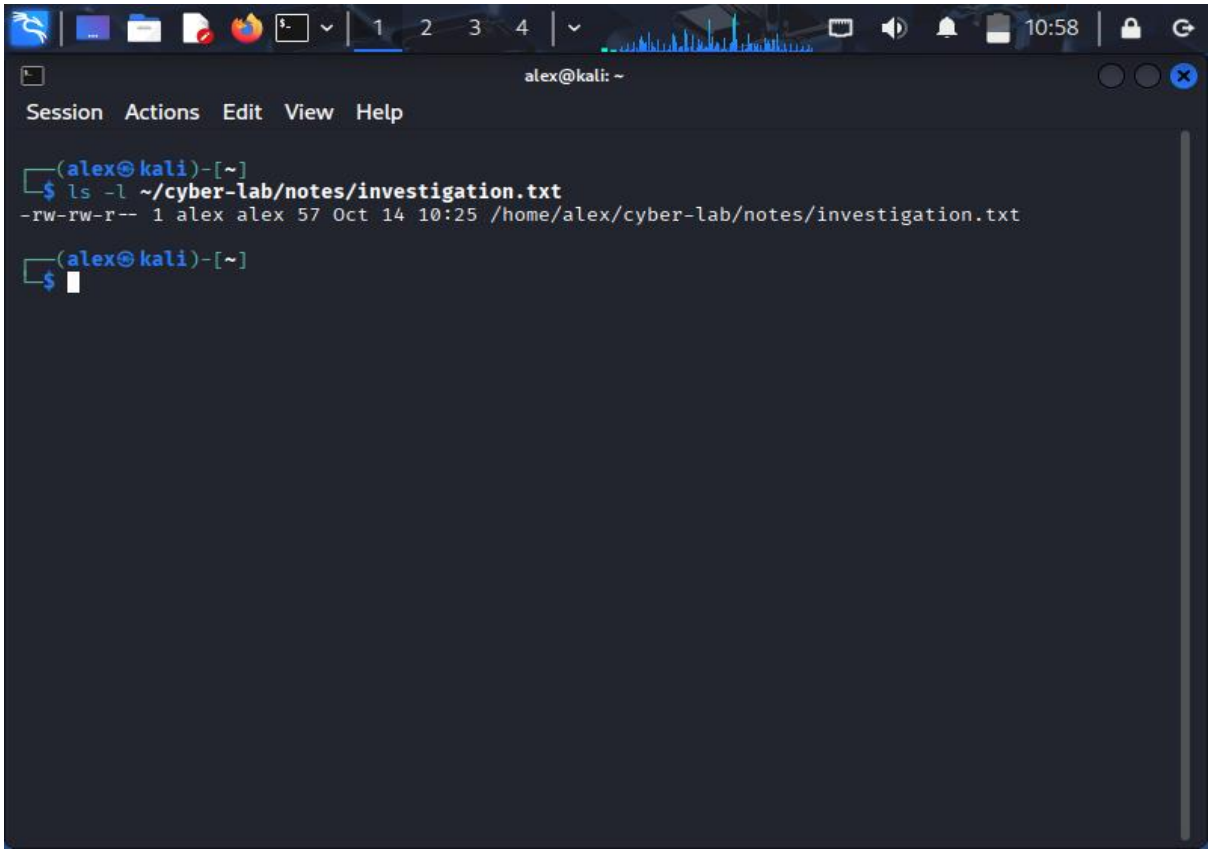
The **echo** command, together with the output redirection operator >**,** was used to create investigation.txt and write the sentence, **"This is my first Linux cybersecurity investigation note."**, into the file.

The **cp** command was used to create a backup copy of the original file. The mv command was then used to rename the **backup to old_investigation.txt** while preserving the original file.

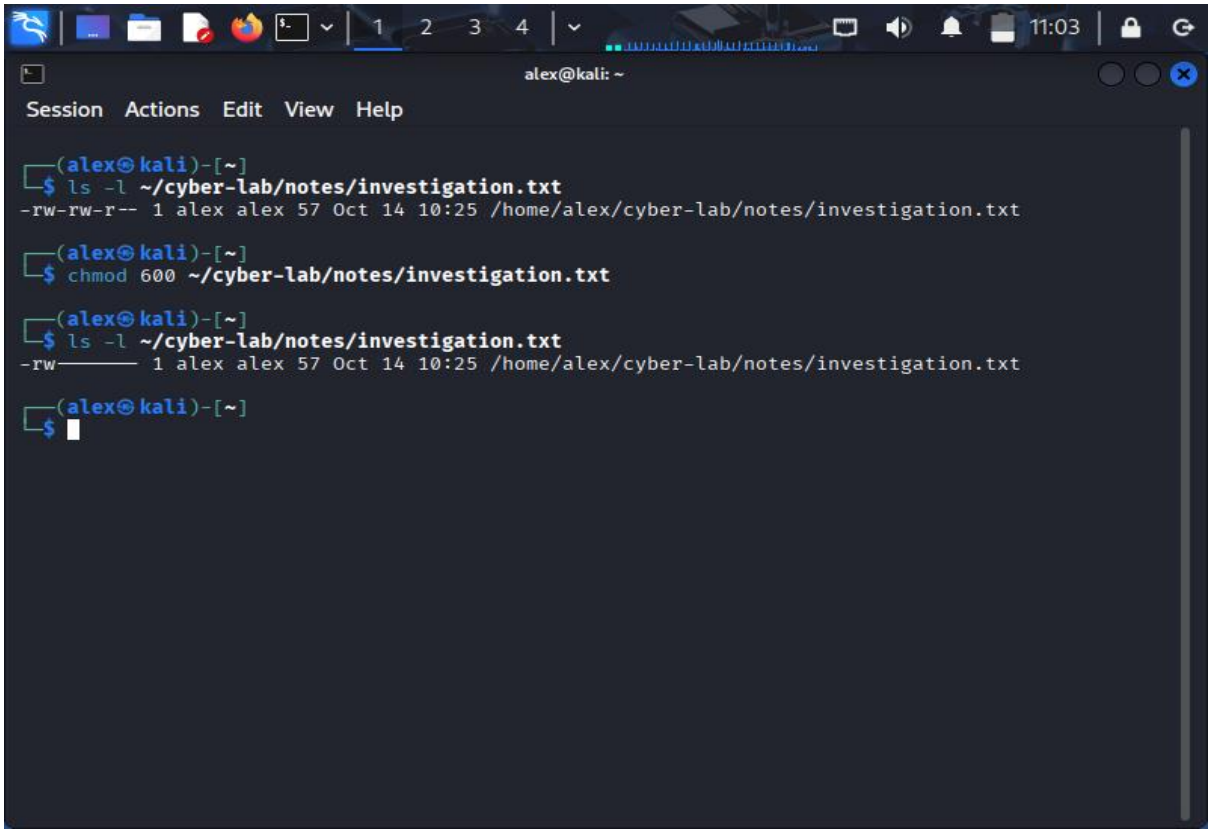
The **ls -R** and **ls -l** commands were used to verify the directory structure and confirm that both files existed. The cat command was used to display the contents of the original investigation note.

TASK 5: PROTECT A SENSITIVE FILE WITH PERMISSIONS

Original permissions of investigation.txt



Changing and verifying file permissions



Commands Used and Explanation

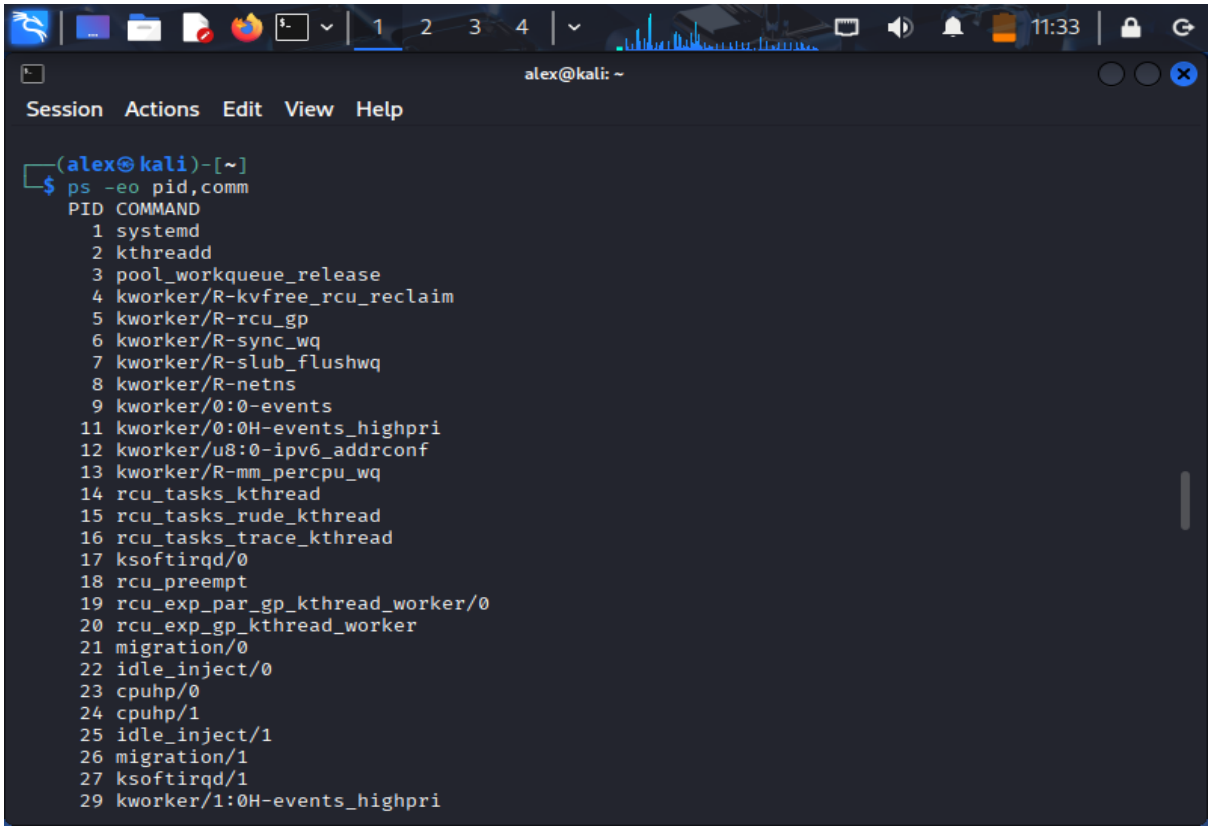
The **ls -l** command was used to inspect the original permissions of investigation.txt. This displayed the file's permission notation before the security change.

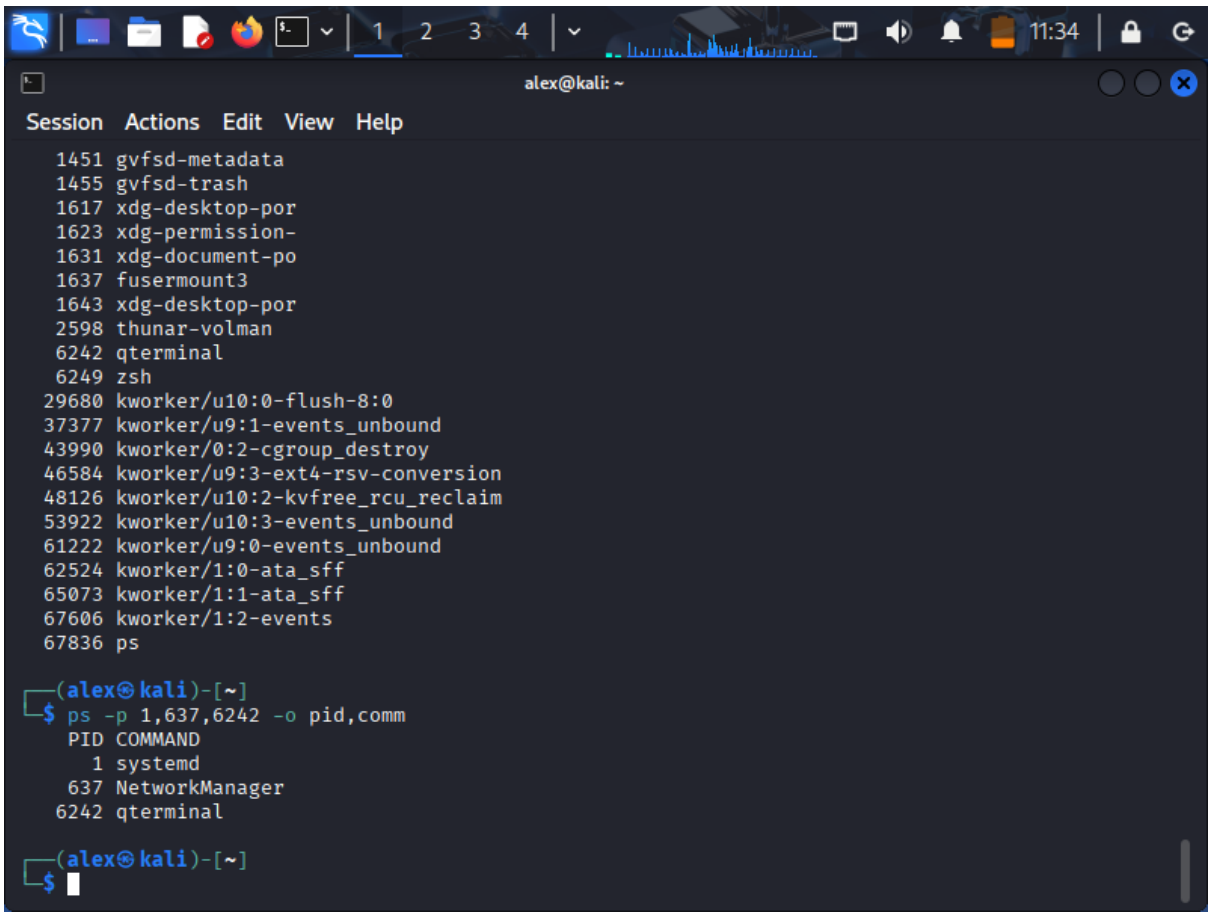
The **chmod 600** command was then used to restrict access to the file. The resulting permission notation, **-rw-------,** indicates that the owner can read and write the file, while the group and other users have no permissions granted through the file's permission bits.
In Linux, read permission **r** allows a user to view a file's contents, write permission **w** allows a user to modify its contents, and execute permission **x** allows a file to be run as a program or script, subject to other system controls.

The numeric permission notation assigns **4** to read, **2** to write, and 1 to execute. Therefore, **6** represents read and write permissions, while **0** represents no permissions. The notation **600** grants read and write access to the owner and no permissions to the group or other users.

TASK 6: INSPECT RUNNING PROCESSES

Running processes in Kali Linux





Processes Identified

The **ps -eo pid,comm,%cpu,%mem** command was used to display a snapshot of running processes on my Kali Linux system. It shows each process **ID**, command name, **CPU** usage, and memory usage.

The three processes identified from my terminal output were:

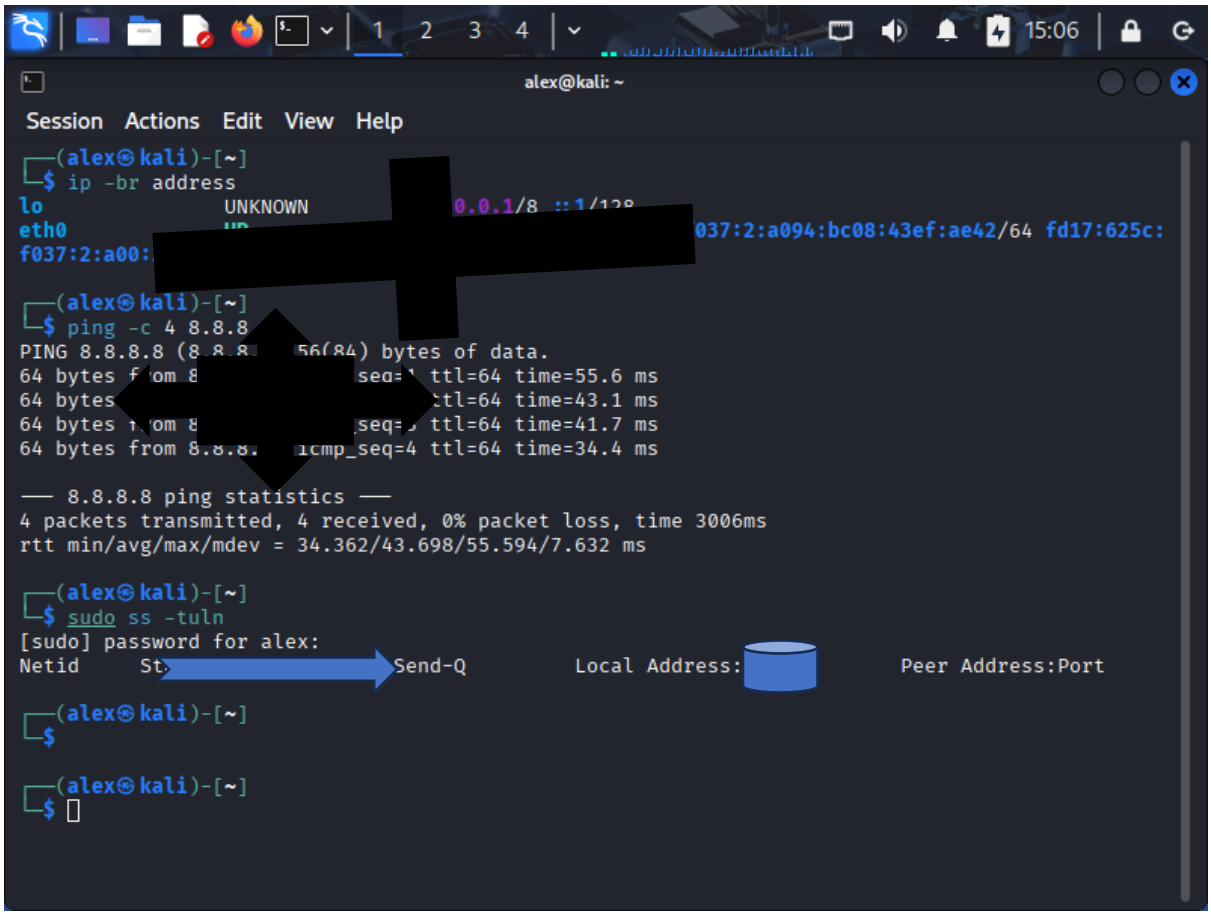
1. **systemd (PID 1):** The first userspace process started by Linux. It manages system services and helps initialize the operating system.
2. **NetworkManager (PID 637):** Manages network interfaces and network connections on the Linux system
3. **qterminal (PID 6242):** A graphical terminal emulator that allows users to interact with the Linux shell by entering commands.

Cybersecurity Importance

Inspecting running processes is useful during a cybersecurity investigation because it helps an analyst understand what programs are active on a system. Unexpected processes may indicate suspicious activity, unauthorized software, or potentially malicious programs. Process information can also help investigators identify resource-intensive applications and determine which processes may require further examination. However, a process should not be considered malicious solely because its name is unfamiliar; additional evidence is needed to assess its behavior.

TASK 7: INVESTIGATE NETWORK INFORMATION

Determine the VM's IP Address



Command used:

ip -br address

Result:

The command displayed two network interfaces:

- **lo** The loopback interface, with IPv4 address **127.0**
- **eth0** The active network interface, with IPv4 address **10.0.2. 4.**

The IPv4 address assigned to my Kali Linux virtual machine's active network interface is **1 5**, with a subnet prefix of /24. The interface also has IPv6 addresses assigned.

Test External Connectivity

Command used:

ping -c 4 8.8.8.8

Result:

- Packets transmitted: 4
- Packets received: 4
- Packet loss: 0%
- Average round-trip time: 43.698 milliseconds

The test was successful because all four packets received replies from **8.8.8.8**, Google's public DNS server. This confirms that the virtual machine could reach the external system using ICMP at the time of testing.

Inspect Listening Ports and Services

Command used:

sudo ss -tuln

Result:

The command displayed the column headings but no listening socket entries.

Therefore, no listening TCP or UDP sockets were shown by this command at the time of inspection.

The command uses -t to display TCP sockets, -u for UDP sockets, -l for listening sockets, and **-n** to display numerical addresses and port numbers.

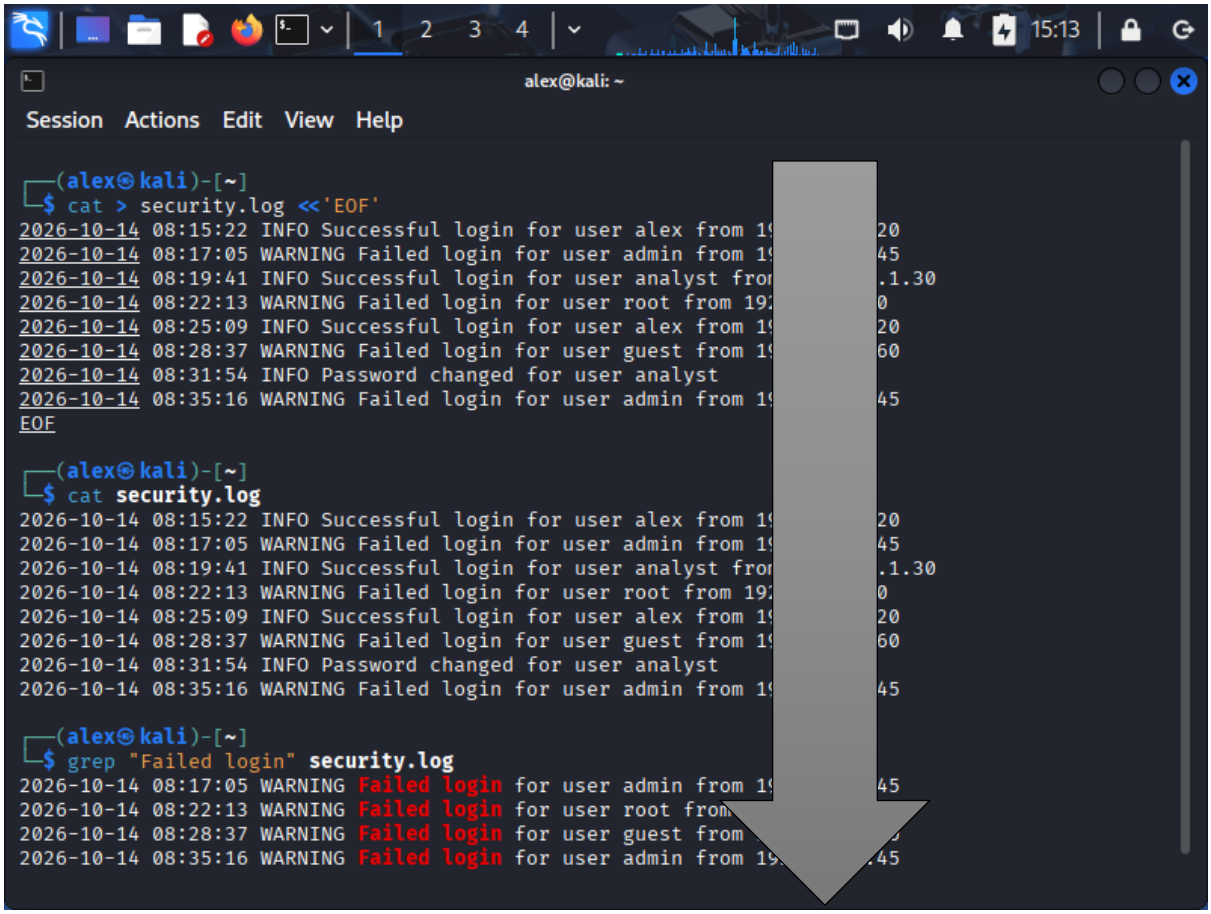
Importance to a Security Analyst

IP addresses: IP addresses help security analysts identify network interfaces and understand how devices communicate. They are useful when investigating network traffic, identifying systems, and troubleshooting network configurations.

External connectivity: Testing connectivity helps analysts determine whether a system can communicate with external hosts. It can help identify network failures, routing issues, or restrictions that may affect security monitoring and system updates.

Listening ports and services: Listening ports help analysts identify services that may accept incoming network traffic. Unnecessary or exposed services can increase a system's attack surface, so analysts inspect them to identify potential security risks and reduce unnecessary exposure.

Task 8: Search a Security Log



2. Create and Display the Log File

I created a file named security.log containing eight sample security events. The events included successful logins, failed login attempts, and a password change.

Command used:

cat security.log

This command displayed the complete contents of the log file.

Search for Failed Login Events

Command used:

grep "Failed login" security.log

The grep command searches text for a specified pattern. In this task, it displayed only the lines containing Failed login, leaving the original file unchanged.

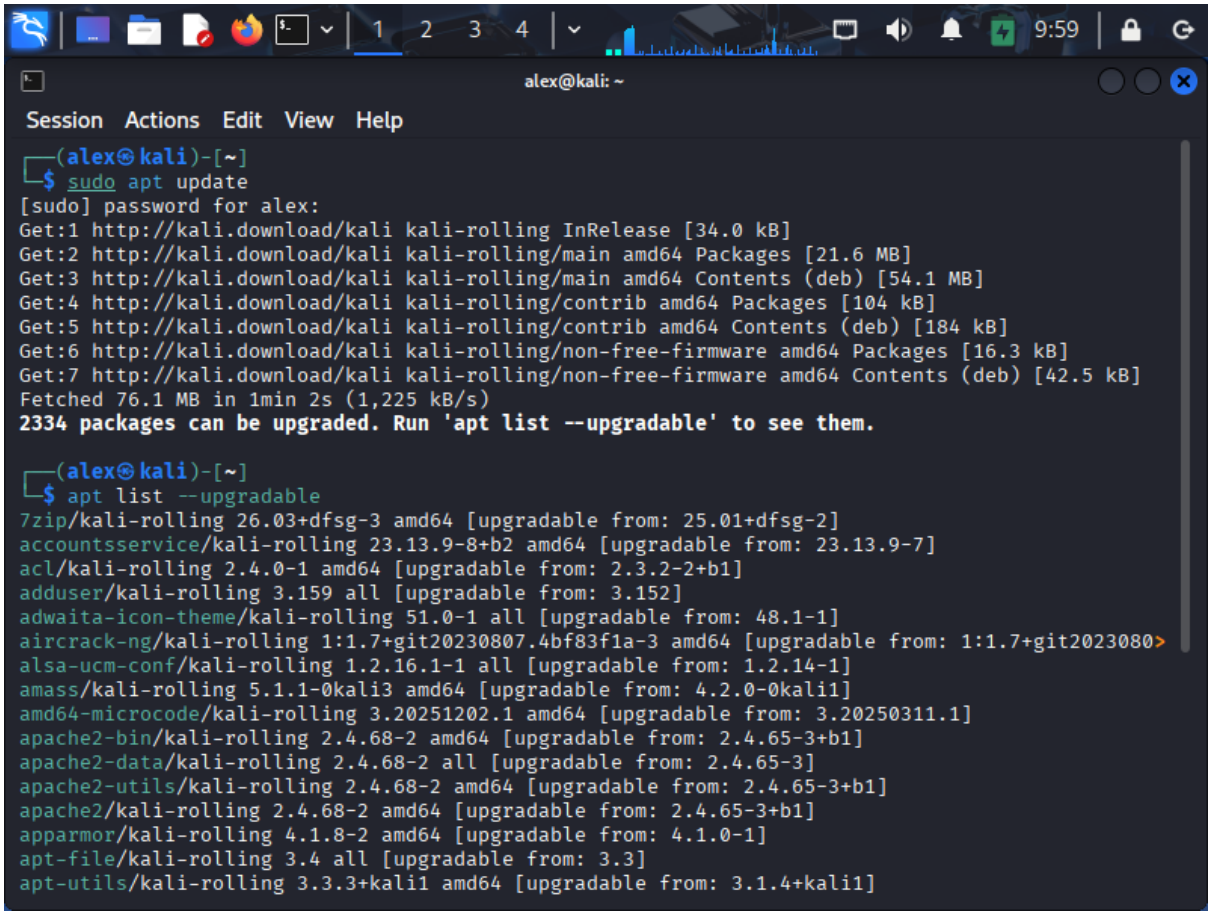
The search returned four failed login entries involving the users admin, root, and guest.

Importance of Log Searching in Cybersecurity

Log searching helps security analysts identify suspicious activities, investigate failed authentication attempts, and recognize possible unauthorized access attempts. Repeated failed logins against an account may indicate password guessing or a brute-force attack. Analysts

can use timestamps, usernames, and source IP addresses in logs to investigate the activity and determine whether further action is required.

Task 9: Check for Available Software Updates



Command Used

The following command was used to refresh the package information:

sudo apt update

This command updates the local package index using the configured software repositories. It does not install or upgrade software.

The following command was used to list installed packages with available upgrades:

apt list --upgradable

This command displays packages for which newer versions are available.

Findings

I ran the command **sudo apt update** to refresh the local package index. The terminal reported that 2,333 packages could be upgraded. However, Kali Linux could not connect to **http.kali.org on port 80**, and some package index files failed to download. Therefore, the

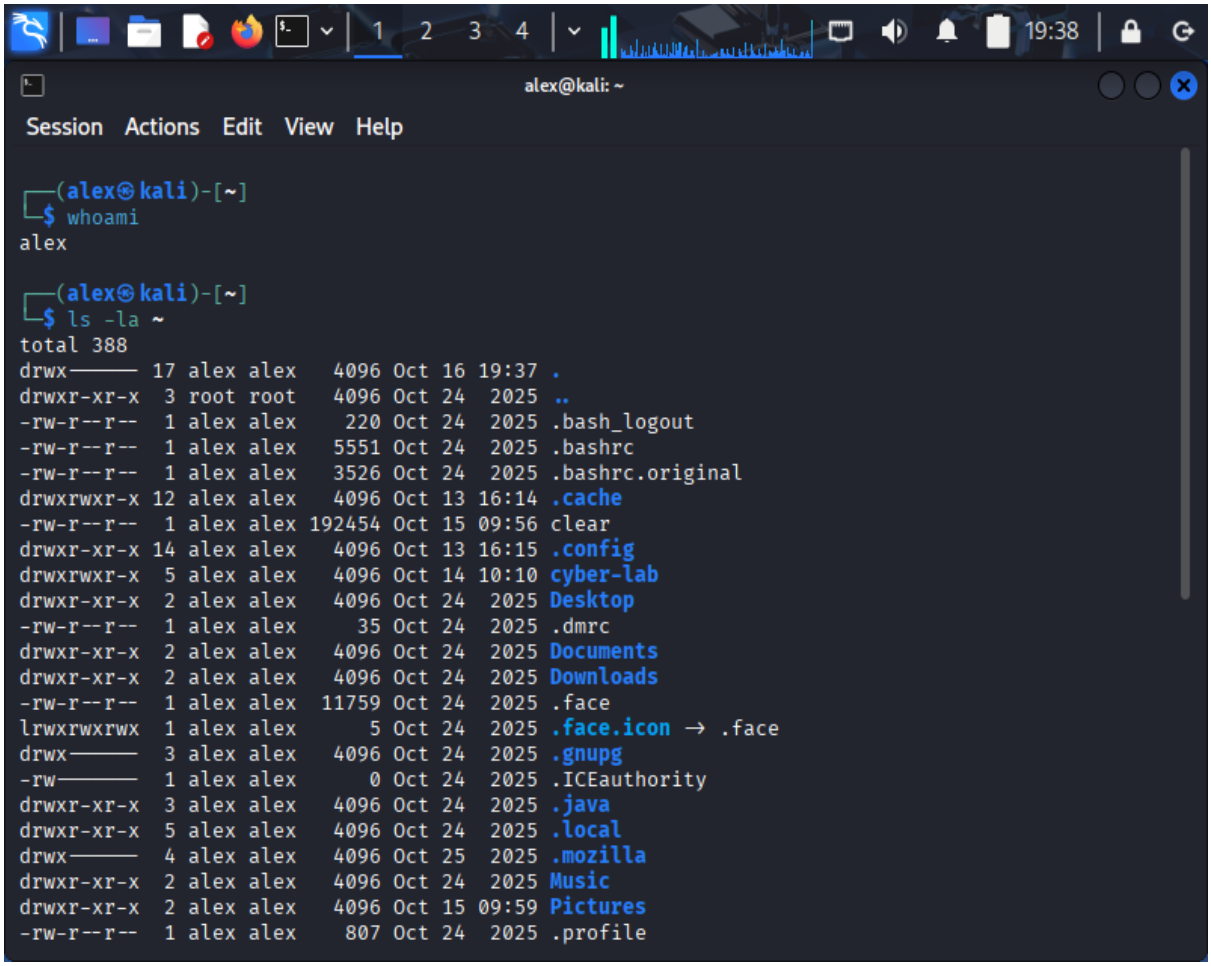
reported number reflects the locally available package information and has not been confirmed as current.

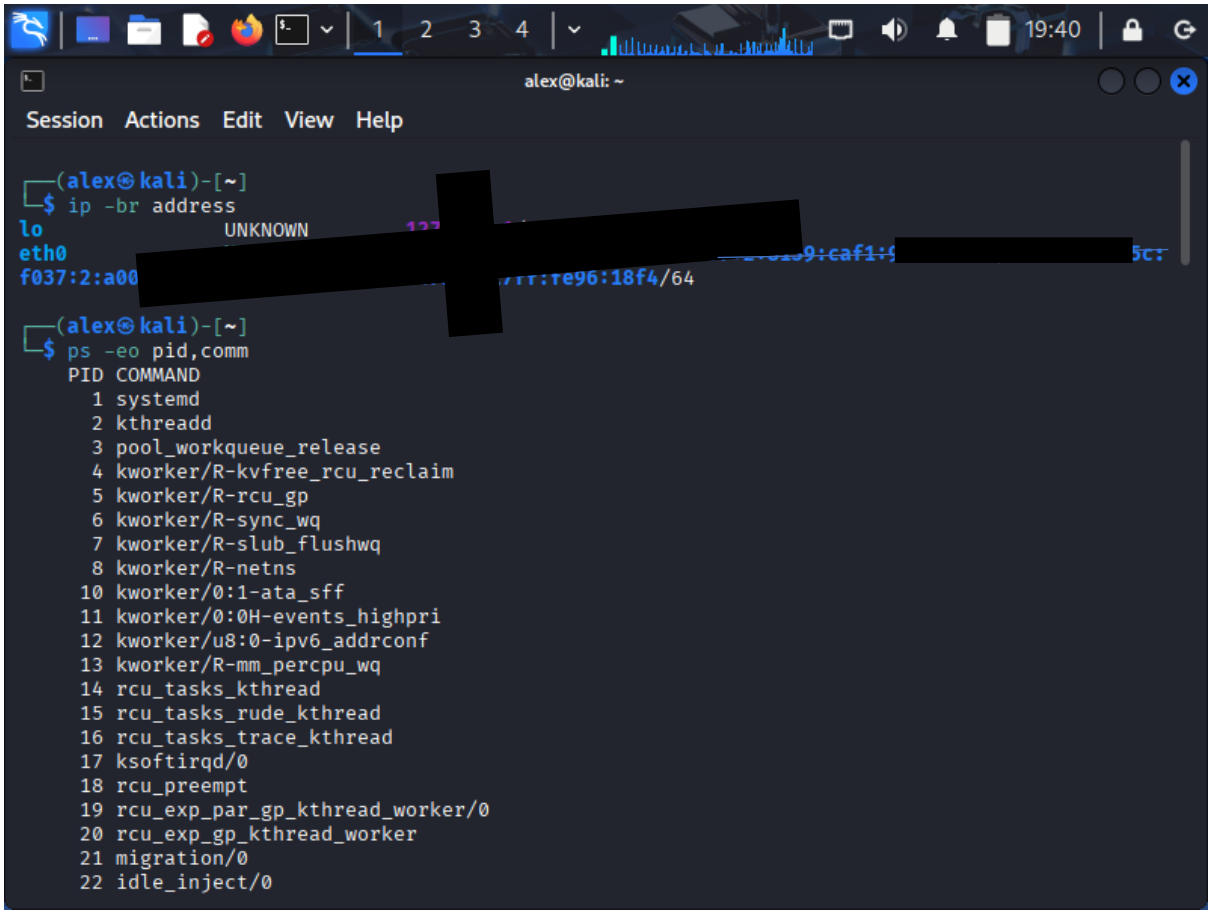
No packages were installed or upgraded during this check.

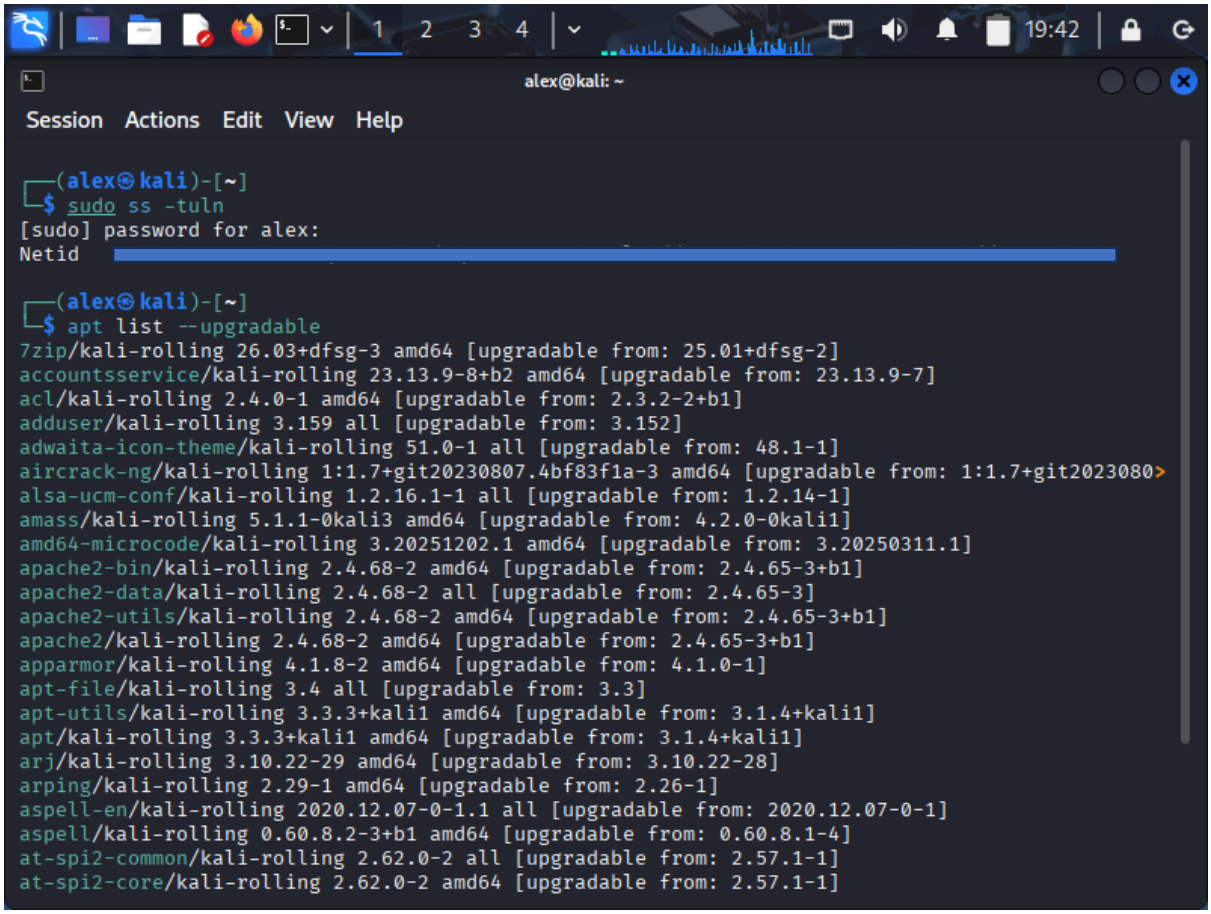
Cybersecurity Importance

Regular software updates and security patches help fix known vulnerabilities, improve system stability, and reduce exposure to attacks. Security analysts should check for updates regularly and apply appropriate patches after considering their relevance and potential impact. Reliable access to software repositories is important because it allows systems to obtain current package information and security fixes.

Task 10: Independent Security Inspection Challenge




1 2 3 4
19:40
alex@kali: ~
Session Actions Edit View Help
(alex㉿kali)-[~]
$ ip -br address
lo UNKNOWN
eth0
f037:2:a00
[redacted]
:re96:18f4/64
(alex㉿kali)-[~]
$ ps -eo pid,comm
PID COMMAND
1 systemd
2 kthreadd
3 pool_workqueue_release
4 kworker/R-kvfree_rcu_reclaim
5 kworker/R-rcu_gp
6 kworker/R-sync_wq
7 kworker/R-slub_flushwq
8 kworker/R-netns
10 kworker/0:1-ata_sff
11 kworker/0:0H-events_highpri
12 kworker/u8:0-ipv6_addrconf
13 kworker/R-mm_percpu_wq
14 rcu_tasks_kthread
15 rcu_tasks_rude_kthread
16 rcu_tasks_trace_kthread
17 ksoftirqd/0
18 rcu_preempt
19 rcu_exp_par_gp_kthread_worker/0
20 rcu_exp_gp_kthread_worker
21 migration/0
22 idle_inject/0


1 2 3 4
19:42
alex@kali: ~
Session Actions Edit View Help
(alex㉿kali)-[~]
$ sudo ss -tuln
[sudo] password for alex:
Netid
(alex㉿kali)-[~]
$ apt list --upgradable
7zip/kali-rolling 26.03+dfsg-3 amd64 [upgradable from: 25.01+dfsg-2]
accountsservice/kali-rolling 23.13.9-8+b2 amd64 [upgradable from: 23.13.9-7]
acl/kali-rolling 2.4.0-1 amd64 [upgradable from: 2.3.2-2+b1]
adduser/kali-rolling 3.159 all [upgradable from: 3.152]
adwaita-icon-theme/kali-rolling 51.0-1 all [upgradable from: 48.1-1]
aircrack-ng/kali-rolling 1:1.7+git20230807.4bf83f1a-3 amd64 [upgradable from: 1:1.7+git2023080>
alsa-ucm-conf/kali-rolling 1.2.16.1-1 all [upgradable from: 1.2.14-1]
amass/kali-rolling 5.1.1-0kali3 amd64 [upgradable from: 4.2.0-0kali1]
amd64-microcode/kali-rolling 3.20251202.1 amd64 [upgradable from: 3.20250311.1]
apache2-bin/kali-rolling 2.4.68-2 amd64 [upgradable from: 2.4.65-3+b1]
apache2-data/kali-rolling 2.4.68-2 all [upgradable from: 2.4.65-3]
apache2-utils/kali-rolling 2.4.68-2 amd64 [upgradable from: 2.4.65-3+b1]
apache2/kali-rolling 2.4.68-2 amd64 [upgradable from: 2.4.65-3+b1]
apparmor/kali-rolling 4.1.8-2 amd64 [upgradable from: 4.1.0-1]
apt-file/kali-rolling 3.4 all [upgradable from: 3.3]
apt-utils/kali-rolling 3.3.3+kali1 amd64 [upgradable from: 3.1.4+kali1]
apt/kali-rolling 3.3.3+kali1 amd64 [upgradable from: 3.1.4+kali1]
arj/kali-rolling 3.10.22-29 amd64 [upgradable from: 3.10.22-28]
arping/kali-rolling 2.29-1 amd64 [upgradable from: 2.26-1]
aspell-en/kali-rolling 2020.12.07-0-1.1 all [upgradable from: 2020.12.07-0-1]
aspell/kali-rolling 0.60.8.2-3+b1 amd64 [upgradable from: 0.60.8.1-4]
at-spi2-common/kali-rolling 2.62.0-2 all [upgradable from: 2.57.1-1]
at-spi2-core/kali-rolling 2.62.0-2 amd64 [upgradable from: 2.57.1-1]

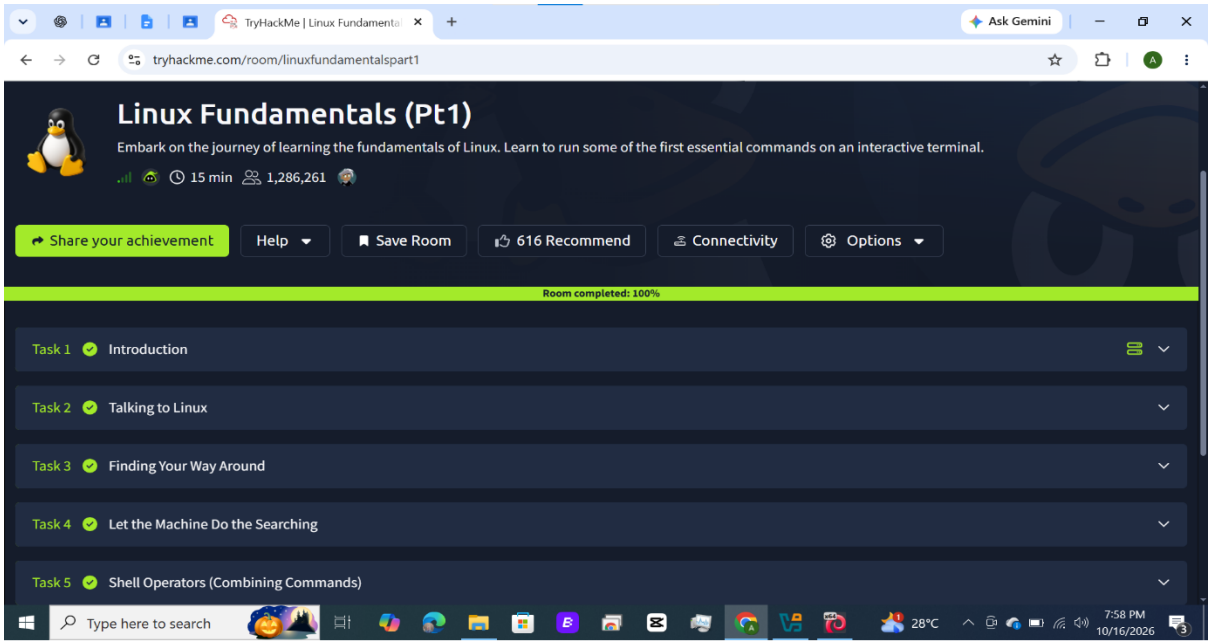
My inspection plan

S/N	Task	Command	Meaning
1	User identity	whoami	Find out which user account is active in the terminal.
2	Filesystem inspection	ls -la ~	Inspect the home directory, including hidden files and their permissions.
3	Network configuration	ip -br address	Identify active network interfaces and assigned IP addresses.
4	Running processes	ps -eo pid,comm	See which processes are running.
5	Listening network services	sudo ss -tuln	Check whether TCP or UDP sockets are listening for connections.
6	Software update status	apt list --upgradable	Check whether locally known package information indicates available software updates.

Task 11: TryHackMe Linux Fundamentals Practice

Room completed: Linux Fundamentals Part 1

Evidence of progress



Three Things I Learned

1. Understanding the Linux filesystem

I learned that Linux organizes files and directories in a hierarchical structure. I practiced navigating between directories using **cd**, checking my current location using **pwd**, and listing files using ls. These commands help me locate and understand files within a Linux system.

2. Working with files and directories

I learned how to view file contents using cat and manage files and directories using commands such as **mkdir** and **rm**. I also understood the importance of checking the correct file or directory before deleting anything to avoid accidentally losing important information.

3. Using Linux commands for cybersecurity tasks

I learned that the Linux terminal allows users to interact directly with the operating system and inspect system information. These skills are useful in cybersecurity because an analyst can use Linux commands to examine files, investigate system activity, and search for information relevant to a security incident.

Task 12: Final Reflection

Linux knowledge is useful to someone working in cybersecurity because it helps them understand how computer systems operate and how to identify possible security problems. Many cybersecurity tools and servers use Linux, so knowing how to work with its commands is an important skill.

One important area I learned about is **users and file permissions**. Commands such as whoami, ls -l, and chmod help me identify the current user and understand who can read, modify, or execute files. This is important because incorrect permissions can expose sensitive information to unauthorized users.

Another area is **processes and networking**. The ps command helps me inspect running processes, while ip and ss provide information about network interfaces and listening ports. These commands can help a security analyst investigate unusual programs, check network configurations, and identify services that may need further inspection.

I also learned about **security logs and software updates**. Using grep to search logs makes it easier to find failed login attempts that could indicate suspicious activity. Checking for software updates helps identify packages that may need security patches to reduce exposure to known vulnerabilities.

Overall, Linux knowledge provides a practical foundation for cybersecurity investigations. It helps analysts examine systems, understand evidence, troubleshoot problems, and make better decisions when protecting computers and networks from potential threats.